

SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense

Editable student template for application-based cloud virtualization and SDDC/cloud operations defense



Assessment Metadata

Course Code	CSA1517
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense
Submission Type	Individual digital case analysis and infrastructure defense based on the same capstone title used in CO1, CO2 AT1 and CO2 AT2
Faculty	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Submission Mode	Completed DOCX template and GitHub repository link through Google Classroom

How This Template Works

- Use the same capstone title used in CO1, CO2 AT1 and CO2 AT2.
- Do not recalculate all VM sizing or repeat all configuration screenshots. Summarize the final AT1/AT2 evidence and use it for design defense.
- Answer all five questions with application-oriented reasoning connected to the capstone product.
- Include sustainable cloud operation decisions wherever relevant: right-sized resources, log retention, storage lifecycle, scheduled shutdown, API-call reduction and VM/container/serverless comparison.
- Paste the final topology diagram in the given response area. The sample diagram is only a structure; redraw it for the student's own product.
- Upload this completed document and supporting evidence to GitHub, then submit the repository link through Google Classroom.

Assessment Focus

AT1 answered what VM is required. AT2 showed configuration or simulation evidence. AT3 asks whether the student can defend the complete infrastructure design under application, security, operations and failure conditions.

Student and Capstone Details

Student Name	Vallapu Mohan Kumar
--------------	---------------------

Register Number	192524225
Class / Slot	Slot B
Capstone Title	Food Trace Cloud Supply Chain Tracker With QR Provenance and Spoilage Prediction
CO1 Evidence Link	
CO2 AT1 VM Recommendation	vCPU: 2 RAM: 4 GB Storage: 50GB VM Class: ____
CO2 AT2 Evidence Link	
GitHub Repository Link	
Submission Date	

Evidence Carry-Forward Summary

Evidence Source		Student Entry	How It Supports AT3 Defense
CO1 concept map / presentation		Cloud architecture and project planning	Defines the overall cloud solution
CO2 AT1 sizing workbook		VM sizing (2 vCPU, 4 GB RAM, 50 GB SSD)	Justifies infrastructure sizing
VM calibration sheet		Azure/VirtualBox configuration	Confirms selected VM resources
CO2 AT2 practical evidence		VM deployment, Docker, security	Validates practical implementation
Capstone architecture decision		Multi-tier cloud architecture	Supports deployment and scalability

Q1: Virtualization Value and Capstone Cloud Deployment Summary

Explain your capstone product, expected workload, cloud need and deployment context. Analyze how virtualization improves cost efficiency, resource utilization, workload isolation, infrastructure flexibility and cloud deployment readiness for your product.

Design Point	Student Response
Capstone product and users	Food Trace is a cloud-based application that enables users to scan QR codes on food products to view supply chain information and receive spoilage predictions. The system is intended for consumers, retailers, suppliers, and administrators.
Expected workload	Around 500–1000 users per day with approximately 720 requests per minute during peak usage.
Cloud need	Cloud infrastructure provides scalable computing, secure storage, AI-based spoilage prediction, and reliable access from anywhere.

Virtualization value	Virtualization allows multiple isolated virtual machines to share physical hardware efficiently while improving flexibility and scalability.
Cost and utilization benefit	Right-sized virtual machines reduce infrastructure costs while maximizing CPU, memory, and storage utilization.
Isolation and flexibility benefit	Each application component operates independently, improving security, maintenance, and fault isolation.
Case Analysis Response: Write 8-12 technical lines connecting virtualization value to your capstone product. Food Trace uses cloud virtualization to provide a scalable and reliable infrastructure for QR-based food tracking and spoilage prediction. Virtual machines isolate application services, preventing failures in one service from affecting others. Cloud resources can be increased or decreased according to user demand, reducing unnecessary costs. Virtualization enables faster deployment, simplified maintenance, and secure resource allocation. The cloud infrastructure supports machine learning, cloud databases, and secure storage services. Load balancing improves application performance during peak traffic. Virtualization also supports backup, disaster recovery, and high availability. Overall, virtualization provides a secure, flexible, and cost-effective environment for deploying the FoodTrace application.	

Q2: Rapid Provisioning, Testing and Virtual Machine Lifecycle Case

Using your CO2 AT1 VM recommendation and CO2 AT2 practical evidence, explain how virtual machines, VM images, templates, snapshots, clones or containers support rapid provisioning, testing, deployment, rollback and maintenance for your capstone product.

Lifecycle Element	AT1/AT2 Evidence Used	Capstone Use / Justification
VM image or base environment	Ubuntu Server 22.04 LTS	Provides a stable environment for deploying the FoodTrace application.
Template or repeatable setup	Azure VM template	Enables quick creation of identical virtual machines for development and testing.
Snapshot / rollback	VM snapshot	Restores the VM to a previous working state if configuration changes cause failures.

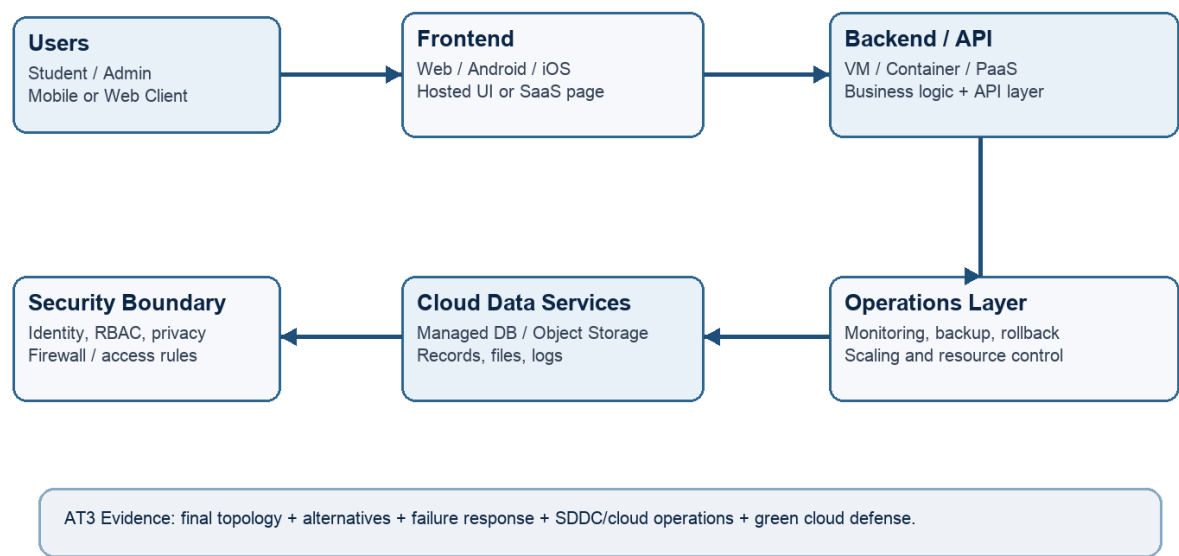
Clone / testing copy	Linked clone	Creates a separate testing environment without affecting the production VM.
Container or VM decision	Docker for backend, VM for infrastructure	Docker provides lightweight deployment while the VM offers secure and isolated infrastructure.
Maintenance and update plan	Scheduled updates and backups	Ensures security patches, application updates, and reliable system performance.
Case Analysis Response: Explain how lifecycle practices reduce deployment delay and risk. Food Trace uses virtual machine lifecycle management to simplify deployment, testing, and maintenance. Ubuntu Server acts as the base environment for hosting the application and its services. VM templates allow identical environments to be created quickly, reducing deployment time. Snapshots provide a recovery point before updates, enabling fast rollback if problems occur. Linked clones create isolated testing environments for new features without affecting the main application. Docker containers are used for backend services because they are lightweight and portable, while the VM provides secure infrastructure. Regular backups, scheduled maintenance, and software updates improve reliability and security. These lifecycle practices reduce deployment delays, minimize downtime, and ensure continuous availability of the FoodTrace cloud application.		

Q3: Final Cloud Deployment Topology and SOA/API Service Integration Case

Draw and explain the final deployment topology for your capstone product. Show frontend, backend/API, database, storage, authentication, notification, analytics and any VM/container/managed cloud service components. Explain how SOA/API-based integration connects these modules and supports modular cloud service design.

Capstone Cloud Deployment Topology - Sample Structure

Students should redraw this with their own modules, services, VM/container choices and security boundaries.

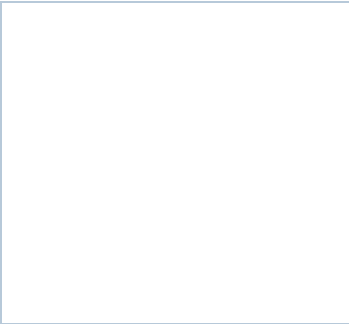


Architecture Component	Technology / Service	VM / Container / Managed Service / SaaS	Integration Method
Frontend	React Web App / Android App	Managed Service	HTTPS REST API
Backend/API	Python Flask API	Docker Container	REST API
Database	Firebase Firestore / Cloud SQL	Managed Service	SQL / Firestore API
Storage	Google Cloud Storage	Managed Service	Cloud Storage API
Authentication	Firebase Authentication	SaaS	JWT Token
Notification	Firebase Cloud Messaging	SaaS	Push Notification API
Analytics / AI module	Python Machine Learning Model	Docker Container	
Paste or draw the final capstone deployment topology diagram here.			

SOA/API Integration

Explanation: Explain how services communicate and why modular service design is useful.

The FoodTrace application follows a **Service-Oriented Architecture (SOA)** in which each module performs a specific task and communicates through secure REST APIs. The frontend sends QR scan requests to the backend API over HTTPS. The backend validates the user through Firebase Authentication using JWT tokens, retrieves product details from Cloud SQL or Firebase Firestore, and accesses Cloud Storage for QR images and related files. The spoilage prediction module uses a Python machine learning model to estimate the freshness of food products. Firebase Cloud Messaging sends notifications such as spoilage alerts or product updates to users. This modular architecture improves scalability, simplifies maintenance, enhances security, and allows each service to be updated independently without affecting the overall application.



Q4: Design Alternatives, Identity, Access and Federated Cloud Privacy Case

Compare at least two deployment alternatives for your capstone product, such as single VM, multi-tier VM, containerized deployment, managed backend, federated cloud or hybrid cloud model. Justify the selected design by analyzing identity management, role-based access control, federated access, data sharing, privacy protection and security risks.

Deployment Alternative	Advantages	Limitations / Risks	Identity and Privacy Impact
Alternative 1	Easy to deploy, low cost, simple management	Limited scalability and availability	Basic authentication, suitable for small applications
Alternative 2	Highly scalable, portable, faster deployment, easier updates	Requires container management	Strong authentication, role-based access, improved isolation
Selected design	Supports auto-scaling, high availability, secure storage, and easy maintenance	Slightly higher cloud management complexity	Provides secure authentication, RBAC, encrypted communication, and better privacy protection
Security Area		Student Decision	
User roles	Administrator, Supplier, Retailer, Consumer		
Authentication method	Firebase Authentication with JWT Token		
Role-based access control	Each user can access only the features permitted for their role.		
Federated access / partner access	Authorized suppliers and retailers can securely access product information using cloud authentication.		
Sensitive data handled	User information, QR code data, product history, supplier details, spoilage prediction results		

Privacy protection decision	HTTPS encryption, JWT authentication, encrypted cloud database, RBAC, and secure cloud storage
Case Analysis Response: Defend why the selected design is better for identity, access and privacy. The selected containerized cloud deployment with managed services is the most suitable solution for the FoodTrace application. Containers provide portability, scalability, and faster deployment compared to a single virtual machine. Firebase Authentication ensures secure user login and identity verification using JWT tokens. Role-Based Access Control restricts access according to user responsibilities such as administrator, supplier, retailer, and consumer. Sensitive information including product records, supplier details, and spoilage predictions is protected using HTTPS encryption and secure cloud storage. Federated access enables trusted supply chain partners to share data securely without exposing confidential information. This architecture improves security, privacy, scalability, and overall system reliability while supporting future expansion of the FoodTrace cloud platform.	

Q5: Software-Defined Datacenter and Cloud Operations Defense Case

Analyze how your infrastructure will respond to operational and failure scenarios such as VM failure, database growth, traffic spike, credential leakage, service outage, resource contention or cost increase. Explain how SDDC concepts and cloud operations practices such as automated provisioning, software-defined networking, software-defined storage, monitoring, logging, backup, rollback, scaling, resource control and maintenance improve the final infrastructure design. Include Green Computing decisions such as resource right-sizing, storage lifecycle control, log retention, API-call reduction, scheduled shutdown, VM/container/serverless comparison and carbon/energy proxy indicators where relevant.

Operational / Failure Scenario	Expected Impact	SDDC / Cloud Operations Response	Evidence or Control
VM failure	Application becomes unavailable	Deploy a new VM automatically and restore from backup	VM backup and recovery
Database growth	Reduced database performance	Enable database scaling and optimize storage	Cloud SQL auto-scaling and indexing
Traffic spike	Increased response time	Use load balancing and auto-scaling	Load Balancer and Auto Scaling
Credential leakage	Unauthorized system access	Reset credentials, enable MFA, revoke compromised tokens	JWT Authentication and Multi-Factor Authentication
Service outage	Users cannot access services	Failover to backup services and monitor health	High Availability and Cloud Monitoring
Resource contention	Slow application performance	Allocate additional CPU, RAM, and storage	Dynamic resource allocation
Cost increase	Higher cloud expenditure	Right-size resources and shut down idle VMs	Cost monitoring and scheduled shutdown
Operational Readiness Area		Student Plan	
Automated provisioning	Use Infrastructure as Code (IaC) templates to create cloud resources automatically.		
Software-defined networking	Configure virtual networks, firewalls, and security groups using cloud networking services.		
Software-defined storage	Store QR images and product records in scalable cloud storage with automatic replication.		
Monitoring and logging	Use Azure Monitor / Google Cloud Monitoring to track		

	system health and application logs.
Backup and rollback	Schedule daily backups and use VM snapshots for quick recovery.
Scaling and elasticity	Enable auto-scaling to increase or decrease resources based on user traffic.
Resource control and maintenance	Regularly update software, monitor resource usage, and remove unused cloud resources.
Green Computing and sustainability control	Right-size VMs, reduce unnecessary API calls, and shut down idle resources to save energy.
Green Cloud Decision	Student Justification
VM right-sizing from CO2 AT1	Selected 2 vCPU, 4 GB RAM, and 50 GB SSD to avoid resource wastage while meeting application requirements.
VM vs container vs serverless choice	Docker containers are used for backend services because they are lightweight and consume fewer resources than full virtual machines.
Scheduled shutdown / idle-resource control	Non-production virtual machines are automatically shut down after working hours to reduce cloud costs and energy usage.
Log and file retention policy	System logs are retained for 30 days, while old logs are archived or deleted automatically.
Storage lifecycle and backup cleanup	Old backups are removed after the retention period to optimize storage usage.
API-call or polling reduction	Cache frequently accessed data and reduce unnecessary

	polling to minimize API requests and improve efficiency.
Carbon/energy proxy indicator to track	Monitor CPU utilization, VM uptime, energy-efficient resource usage, and cloud cost as indicators of sustainable cloud operations.
Final Infrastructure Defense Statement: State whether the design is ready for prototype, pilot or production. Defend reliability, security, SDDC/cloud operations and Green Computing improvements required next. The proposed cloud infrastructure for FoodTrace Cloud Supply Chain Tracker With QR Provenance and Spoilage Prediction is suitable for a prototype and can be extended to a pilot deployment. The design uses virtual machines, Docker containers, managed cloud services, and cloud storage to provide scalability, reliability, and security. Auto-scaling, load balancing, and continuous monitoring ensure stable performance during peak traffic. Backup, rollback, and disaster recovery strategies improve system availability. Security is strengthened using HTTPS, JWT authentication, role-based access control, and encrypted cloud storage. Green Computing practices such as right-sized virtual machines, scheduled shutdown of idle resources, storage lifecycle management, and reduced API calls help minimize operational cost and energy	

consumption while supporting sustainable cloud infrastructure.

CO2 Reflection and Individual Defense

This section must be individually written. It should show what the student understands from the complete CO2 chain: infrastructure sizing, practical configuration evidence and final operations defense.

Reflection Prompt	Student Response
What is the strongest infrastructure decision in your capstone design?	The strongest infrastructure decision is the use of a containerized cloud architecture with managed cloud services . Docker containers provide lightweight deployment, while managed services such as Cloud SQL, Firebase Authentication, and Cloud Storage improve scalability, security, reliability, and maintenance.
Which CO2 concept became clearer after AT1, AT2 and AT3?	The concepts of virtualization, cloud infrastructure planning, VM sizing, Docker containers, software-defined datacenter (SDDC), cloud security, auto-scaling, monitoring, backup, and disaster recovery became much clearer through the completion of AT1, AT2, and AT3.
Which risk is most serious for your capstone: failure, cost, privacy, scaling or sustainability?	The most serious risk is privacy and data security because the application stores user information, QR code data, supplier details, and product tracking records. Strong authentication, HTTPS encryption, role-based access control, and secure cloud storage are essential to protect sensitive information.
What evidence would you show during viva to defend your infrastructure design?	I would present the VM sizing report from AT1, Azure VM/VirtualBox configuration screenshots, Docker container execution, network configuration, security settings, deployment topology diagram, GitHub repository , and the completed AT2 and AT3 documents to demonstrate the correctness of the infrastructure design.

What will you improve before final capstone deployment?	Before the final deployment, I will implement multi-factor authentication , enable automatic backups , improve the machine learning model for spoilage prediction , configure auto-scaling and load balancing , strengthen monitoring and logging , and optimize cloud resources using Green Computing practices to improve performance, security, and sustainability.
---	--

Student-Facing Assessment Criteria

Criteria	Descriptor
Capstone continuity	The answer uses the same capstone title and connects CO1, CO2 AT1 and CO2 AT2 evidence.
Virtualization and lifecycle reasoning	VM, image, snapshot, clone, container or managed-service decisions are technically justified.
Topology and integration clarity	Architecture diagram and SOA/API explanation show clear module interaction.
Identity, access and privacy	Roles, authentication, RBAC, federated access, sensitive data and privacy decisions are defended.
SDDC/cloud operations	Failure response, monitoring, logging, scaling, backup, rollback and resource control are addressed.
Green Computing defense	Right-sizing, retention, API efficiency, idle-resource control and sustainable cloud choices are explained.
Submission quality	GitHub, template completion, evidence references and individual reflection are complete.

GitHub and Google Classroom Submission

Repository Item	Expected Content
README.md	Capstone title, student details, final VM recommendation, deployment topology summary and AT3 evidence index.
Completed AT3 document	This completed editable DOCX or exported PDF if instructed.
Diagram image	Final topology diagram if created separately.
Evidence references	Links or screenshots from CO1, CO2 AT1 and CO2 AT2 as needed.
Green Computing note	Short README section explaining sustainable cloud choices and resource-control decisions.
Google Classroom	Submit the GitHub repository link through GCR.

Student Assurance

I confirm that this case analysis, design reasoning, topology diagram and final defense statement were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

Student Name	Vallapu Mohan Kumar
Register Number	192524225
Signature	V. Mohan Kumar
Date	13/08/2026

Faculty Verification

Faculty Name	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Constructive Feedback / Remarks	
Strength Observed	
Improvement Suggested	
Signature / Date	